Jefrey Bergl

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Website | GitHub

EDUCATION

University of North Carolina at Chapel Hill Expected Graduation: May 2026
***BS** Computer Science and Applied Mathematics*
Dean's List (5 semesters)
Coursework: Graduate Machine Learning, Graduate Deep Learning, Optimization for ML, Numerical Analysis; Graduate Seminars - Large Language Models, Vision Transformers, Embodied AI

PUBLICATIONS, POSTERS, & PRESENTATIONS

Hofweber, T., **Bergl, J.**, Reyes, I., Sadovnik, A. (2025). "The Black Tuesday Attack: How to Crash the Stock Market With Adversarial Attacks to Financial Forecasting Models." In review, *The Journal of Cybersecurity*. arXiv:2510.18990.

Bergl, J., Reyes, I., Sadovnik, A., Hofweber, T. (2026). "The Black Tuesday Attack." *CoreAI UNC Research Showcase*. [poster]

Bergl, J., Reyes, I. (2025). Adversarial Attacks on Financial Forecasting Models. *AI@UNC*. [slides]
*Selected for Spotlight Presentation.

RESEARCH EXPERIENCE

UNC Chapel Hill Oct 2024 – Present
Undergraduate Research Assistant – Adversarial Attacks on Financial Forecasting Models
Faculty Advisor: Thomas Hofweber
Technical Advisor: Amir Sadovnik, Oak Ridge National Lab

- Helped develop adversarial attack framework, the "Black Tuesday Attack," which disrupts stock indexes.
- Designed and implemented adversarial perturbation framework targeting Convolutional Neural Network and Multi-Layer Perceptron forecasting architectures.
- Received $3,000 research stipend.

INDEPENDENT RESEARCH PROJECTS

The Representation-Reasoning Gap: Encoding vs. Reasoning in LMs [GitHub] Jan 2026

- Showed that LMs (GPT-2 Medium, Pythia 1.4B/2.8B) encode spatial position with 98–100% probe accuracy yet achieve only 0–5% behavioral accuracy, revealing a representation-reasoning dissociation.
- Designed a controlled 5×5 grid-world task (2,500 scenarios); trained ridge classifiers on residual-stream activations across all layers and step counts.
- Found position representations consolidate at the final token (+22pp over step tokens), ruling out information loss and localizing the bottleneck to the unembedding layer.

H-JEPA: Learning Hamiltonian Dynamics on Latent Video Embeddings [GitHub] Dec 2025

- Designed a latent world model enforcing Hamiltonian mechanics on frozen V-JEPA 2 embeddings; states evolve via a symplectic Störmer–Verlet integrator with learnable timestep.
- Achieved 1.9× lower prediction MSE than a parameter-matched MLP on 15-step rollouts with 30% fewer parameters (919K vs. 1.3M) and bounded energy drift.
- Conducted ablation studies isolating symplectic integration and energy conservation contributions, revealing a prediction–conservation trade-off across four model variants.

ACTIVITIES

Member: AI@UNC, UNC Quantum Computing Club, UNC Poker Club

TECHNICAL SKILLS

Programming Languages: Python, Java